Cystic Fibrosis (1 of 2)

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Adapted from: Cystic Fibrosis

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentations of cystic fibrosis (CF).
2. Explain the pathophysiology, including the genetic basis, of CF.
3. Interpret clinical findings to properly diagnose CF.

CASE CONNECTION

TF is a 24-year-old female with pulmonary complications from cystic fibrosis who comes to the clinic for follow-up care. She has been hospitalized 15 times for pneumonia, most often due to *Pseudomonas aeruginosa*. She says, "I'm peeing a lot, and I feel thirsty, hungry, and tired." TF's nonfasting glucose and hemoglobin A_{1c} are both elevated. When you tell her she has diabetes and that it is likely related to her cystic fibrosis (CF), she says, "But CF involves the lungs. Lung problems don't cause diabetes, do they?"

What is the potential etiology of TF's hyperglycemia? Consider your answer as you read, and we'll revisit TF at the end of the brick.

GO TO CONCLUSION ↓

What Is Cystic Fibrosis?

Cystic fibrosis (CF) is a multisystem heritable genetic disorder that causes obstructive lung disease, gastrointestinal disease (including malabsorption), and infertility. The respiratory disease is most common and is characterized by thickened, viscous airway mucus leading to obstructed airflow and increased respiratory infections. All of the manifestations of CF are due to **defects in cellular chloride transport**.

Epidemiologically, the prevalence of CF in the United States is about 40,000 patients, with an annual incidence of 1 in 2000 to 4000 live births. Approximately 1000 new cases are diagnosed in the United States every year, mostly as a result of perinatal testing for CF, now routine in many countries, including the United States.

CF is more common in white populations of European ancestry (3-10 fold increase in incidence vs other populations). The median life expectancy for patients with CF has significantly improved over the last six decades—from only a few months to over 40 years—due to better diagnosis and treatment.

CF occurs most often in white populations of European descent.

How Do Patients With Cystic Fibrosis Present (LO #1)?

Cystic fibrosis usually is detected through newborn genetic screening. Once symptoms develop, usually in childhood, there is a general progression of symptoms that can include:

- Respiratory: persistent, productive coughing, wheezing, and dyspnea. There are frequent lung infections. Sinus obstruction leads to postnasal drip, recurrent sinus infections, nasal congestion, headaches, and sleep apnea (frequent awakening from sleep).
- Gastrointestinal: constipation, including thick, sticky meconium (initial stool) in a neonate, creating a blockage in the intestine (meconium ileus). Malabsorption of fat and fat-soluble vitamins may occur later, leading to greasy stools, diarrhea, weight loss, and failure to thrive. Gastroesophageal reflux disease (GERD) is common and causes substernal burning and pain.
- Reproductive: male infertility caused by the congenital absence of the vas deferens. Females have thickened cervical mucus and may have

greater difficulty in conceiving but are usually fertile.

Characteristic pulmonary symptoms of CF include persistent productive coughing, wheezing, shortness of breath, and frequent lung infections.

What Is the Pathophysiology of Cystic Fibrosis (LO #2)?

Cystic fibrosis is caused by **reduced epithelial cell chloride ion transport**. Normally, chloride ion movement is mediated by the cystic fibrosis transmembrane conductance regulator (CFTR), a transport protein found on the apical membrane of epithelial cells. When this ion channel opens, ions flow down their electrochemical gradient (high to low). As a result, in most epithelial cells (eg, lung, pancreas), the CFTR transports chloride (and bicarbonate) out of the cell and into the extracellular fluid, including respiratory and gastrointestinal (GI) secretions. In contrast, in the sweat glands (eg, skin), the transport carries chloride into the cell. We will return to this point when we talk about sweat testing for CF later.

Based on this transport physiology, the CFTR functions to maintain the fluidity of mucus, sweat, saliva, tears, and digestive fluids. The loss of CFTR function in CF causes decreased ion transport and increased viscosity (thickness) of these fluids, resulting in CF symptoms.

Genetics

Cystic fibrosis is caused by a variety of **mutations in the *CFTR* gene**, which is located on the long arm (q) of chromosome 7. CF is an **autosomal recessive** disorder, meaning that both alleles of the *CFTR* gene must be mutated for the phenotypic expression of CF. Researchers have identified more than 2000 variants in the *CFTR* gene, and about 300 are considered pathogenic. All of them reduce or eliminate CFTR protein transport at the cell membrane either through functional changes in the protein or reduced receptor trafficking to the cell membrane.

Cell Biology and Physiology

How do CFTR mutations cause failure of epithelial cell chloride transport? There are multiple mechanisms, depending on the exact genetic mutation. These include:

- CFTR protein processing mutations (misfolding): the most common mutation (90% of patients) results in deletion of **phenylalanine 508**, causing a mutated CFTR protein that misfolds after translation. The CFTR remains sequestered in the rough endoplasmic reticulum rather than migrating to the cell membrane as it should.
- Reduced or absent synthesis (translation) of functional CFTR.
- Functional mutations: CFTR is produced and reaches the cell membrane, but is defective and the ion channel does not open properly or does not conduct ions normally, leading to reduced chloride transport.

The most common type of mutation in CF is deletion of phenylalanine 508 in the cystic fibrosis transmembrane conductance regulator gene, which leads to a protein processing mutation.

Consequences of Decreased CFTR Function

How does the *CFTR* mutation lead to clinical disease in CF? As you may recall from basic physiology, epithelial cells have transporters and channels in their membranes that allow ions and water to pass between the cell's cytoplasm and extracellular environment. This mechanism is critical to normal tissue function. In healthy lungs, GI tract, and pancreas, epithelial cells **secrete chloride ions from the cell interior to the extracellular fluid** using the CFTR channels in the apical cell membrane.

This chloride secretion plays an important normal role in modulating the fluidity of luminal fluids, such as airway mucus in the respiratory tract. Chloride secretion increases the electronegativity of the extracellular fluid, encouraging sodium ions to migrate out of the cell into the extracellular fluid. The increased ions in the extracellular space increase its osmolality, causing water to remain in the airway or lumen rather than entering the cell.

The increased water in the mucus or pancreatic secretion prevents it from becoming too viscous (sticky) (Figure 1, left).

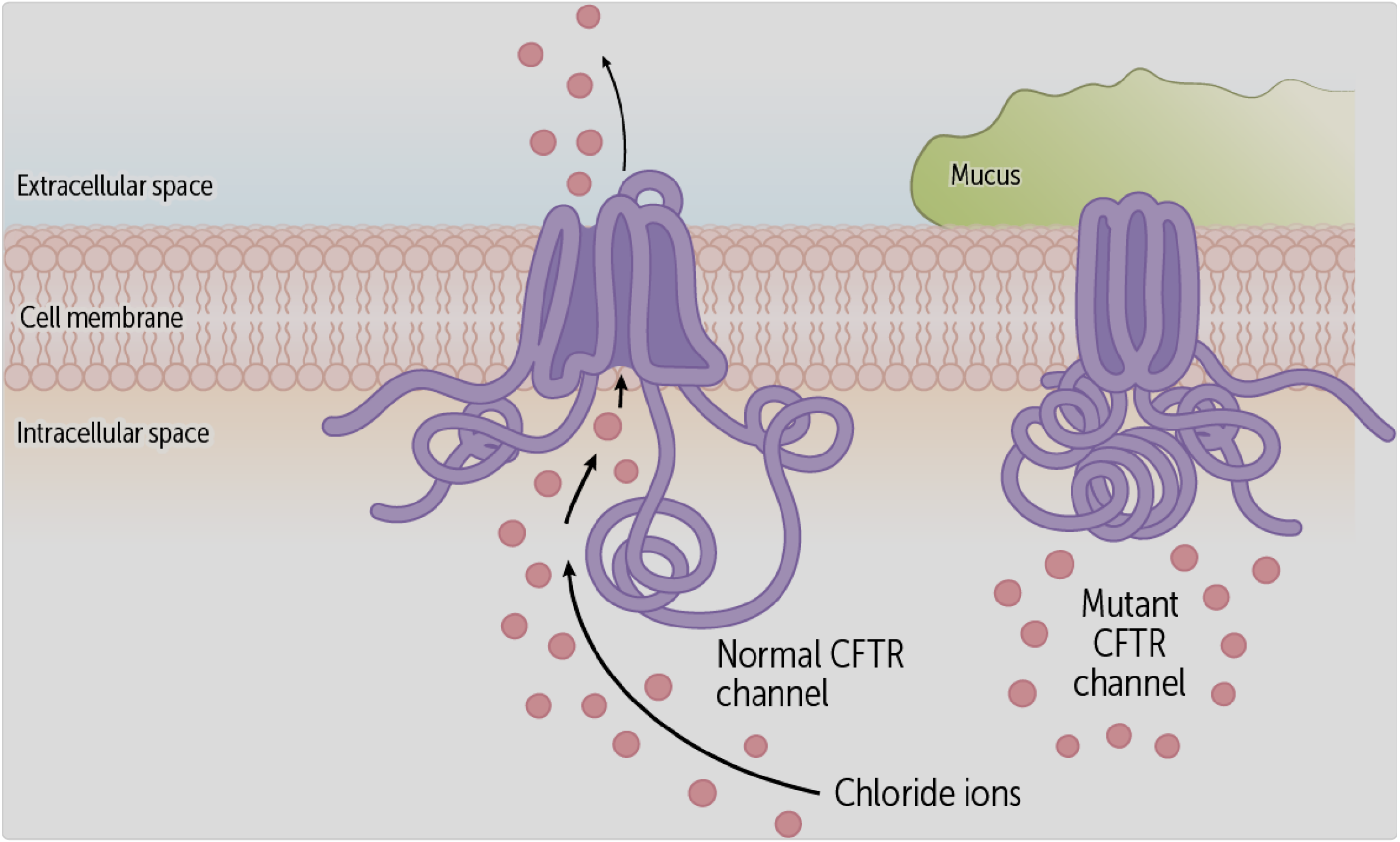

Figure 1

In contrast, in CF the poorly functioning membrane CFTR protein **decreases epithelial cell chloride secretion**, reducing extracellular chloride (see Figure 1, right). With fewer ions in the extracellular fluid, water is more likely to enter the cell, and the remaining mucus is more viscous (Figure 2). This results in the typical CF pathology.

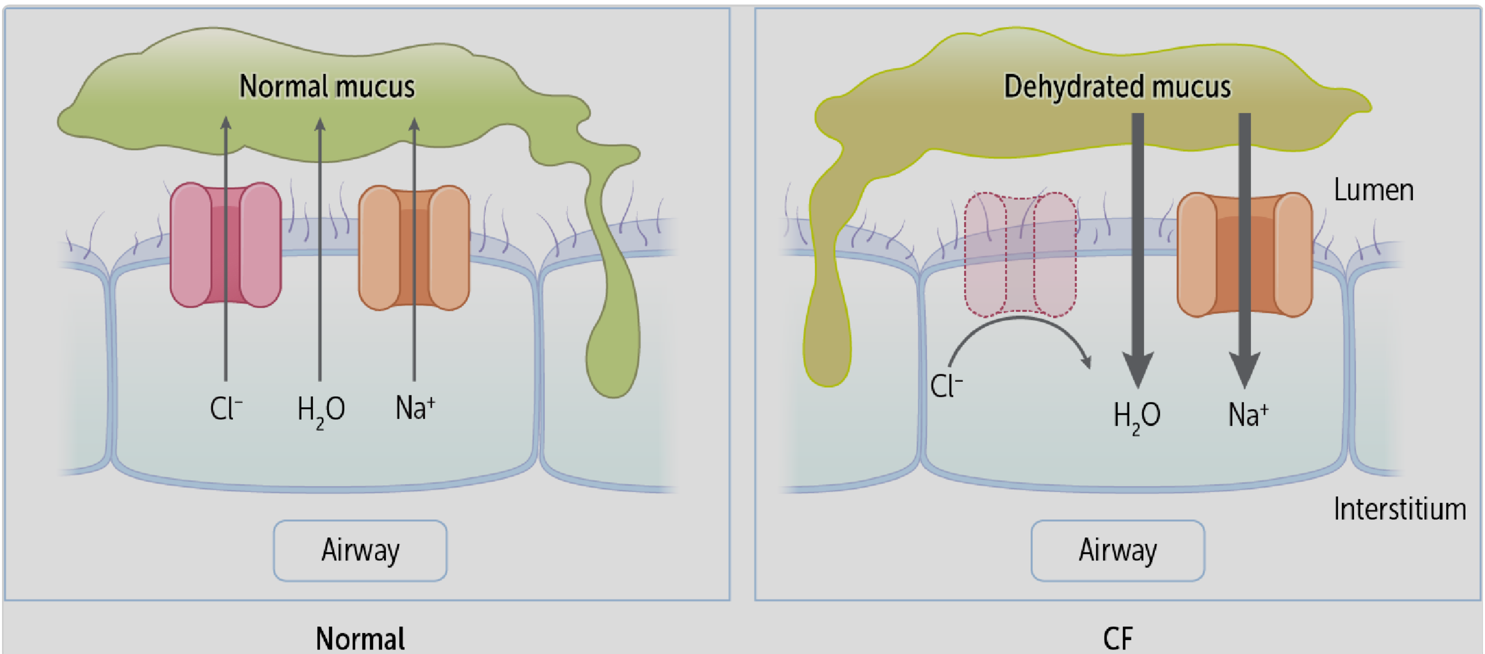

Figure 2

When CFTR is defective, epithelial cell chloride secretion decreases.

Pathologic Consequences of Thick Mucus

In patients with CF, the main consequences of thick mucus are respiratory, gastrointestinal, and reproductive symptoms.

Respiratory Effects. The thick mucus and airway inflammation have several consequences, including the characteristic symptoms of coughing, dyspnea, and wheezing. Some respiratory consequences of CF include:

- **Airway obstruction**: The airways are obstructed, especially during expiration, so CF is classified as an **obstructive pulmonary disease**, in the same class as asthma and chronic obstructive pulmonary disease (COPD). Spirometry will show a reduced FEV_1 and FEV_1/FVC ratio, typical of obstructive diseases.
- **Recurrent infection**: Thick mucus also impairs ciliary function in the respiratory epithelial cells. The cilia normally help remove pathogens from the airways. But when cilia are impaired, patients are more prone to **recurring respiratory infections** such as sinusitis and pneumonia. Childhood infections are often due to *Staphylococcus aureus* and *Haemophilus influenza*. Adolescents and adults, on the other hand, are prone to infection with *Pseudomonas aeruginosa*, a pathogen that colonizes the respiratory tree of patients with CF. The mucoid polysaccharide capsule allows *P. aeruginosa* to establish biofilms, contributing to chronic pneumonia in CF patients that is difficult to treat or eradicate. Allergic bronchopulmonary aspergillosis, a condition characterized by an exaggerated response of the immune system (a

hypersensitivity response) to fungi in the *Aspergillus* genus, occurs most often in people with asthma or cystic fibrosis.

- **Bronchiectasis**: The chronic inflammation of CF makes it the leading cause of bronchiectasis, an irreversible dilation and scarring of the bronchioles and bronchi. Bronchiectasis increases the risk of pneumonia, as well as complications like atelectasis (lung collapse) and hemoptysis.
- **Rhinosinusitis**: In the upper airways, inflammation from chronic rhinosinusitis can cause **nasal polyps and recurrent sinusitis**. Sinus obstruction affects the majority of patients with CF, causing headaches and facial pain.

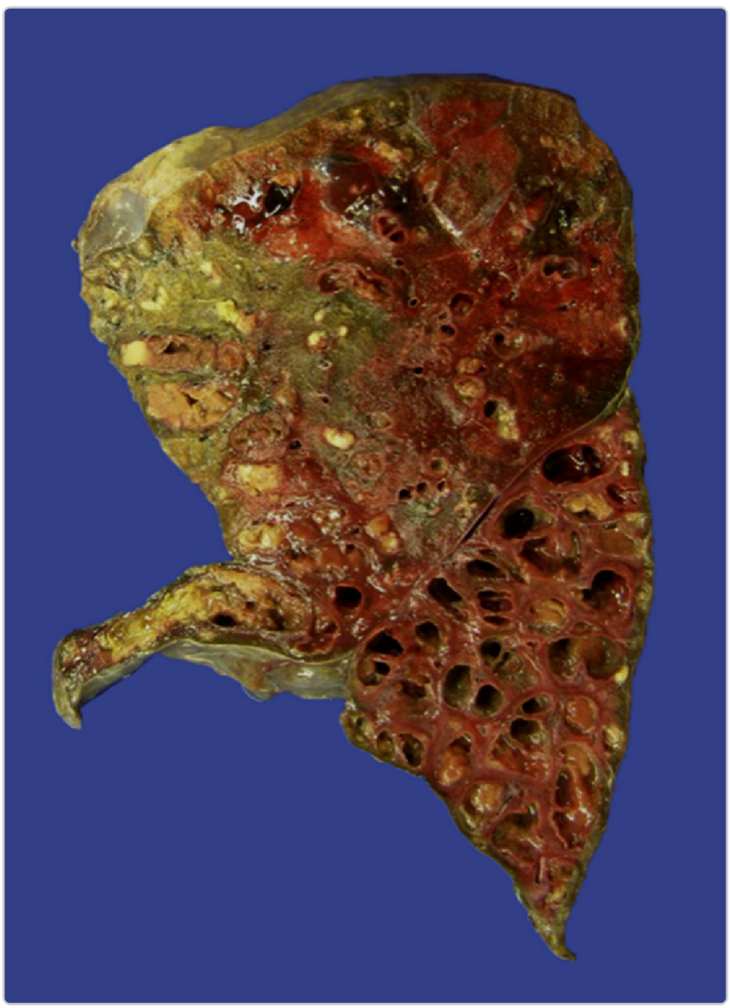

Figure 3. Lungs of a patient who died of cystic fibrosis. There is extensive mucus plugging and dilation of the tracheobronchial tree. The pulmonary parenchyma is consolidated by a combination of both secretions and pneumonia—the green color associated with *Pseudomonas* infection.

Gastrointestinal Effects. These effects are largely due to thickened secretions. Some examples include:

- **Constipation and Ileus**: In neonates, thick, sticky intestinal secretions can cause thick meconium (the neonate's first stool) and create a bowel obstruction known as meconium ileus. In older patients, the same mechanism causes recurrent constipation.
- **Exocrine pancreas insufficiency and malabsorption**: Thick mucus may obstruct the pancreatic ducts. As a result, the pancreas is unable to release enzymes into the gut for digestion and absorption of macromolecules (ie, fats, proteins, carbohydrates), a condition known as **pancreatic insufficiency**. Fatty acids, as well as the fat-soluble vitamins A, D, E, and K, are poorly absorbed. Malabsorption can cause **malnutrition and failure to thrive** in infants, along with fat-soluble vitamin deficiencies. This malabsorption can lead to **steatorrhea** (loose, fatty stools) because mucus and fat nutrients remain in the gut lumen. Patients may develop bone disease due to malabsorption of vitamin D.
- **Secondary diabetes mellitus**: Over time, blockage of the pancreatic ducts leads to the replacement of tissue with fibrosis and adipose tissue. This will eventually impair both pancreatic exocrine function (digestion) as well as endocrine function (insulin production). A consequence is **cystic fibrosis–related diabetes mellitus (CFRD)**, which shares features of both type 1 and type 2 diabetes. CFRD is common in CF patients, affecting approximately 20% of adolescents and up to 50% of adults with CF.
- **Liver disease**: CF also causes liver disease, due to reduced flow of thickened bile (cholestasis), eventually leading to **biliary cirrhosis**, characterized by scarring of the liver, progressive liver failure, and

increased portal vein pressure (portal hypertension). Patients with CF also have an increased incidence of gallstones and cholecystitis (gall bladder inflammation), leading to abdominal pain.

Reproductive and Fertility Effects. Up to 98% of males with CF are **infertile** because they are born with an absent or obstructed vas deferens, which normally transports sperm from the epididymis to the ejaculatory ducts. This is likely because CFTR plays an important role in the formation of the vas deferens during development.

Females with CF, on the other hand, may have trouble getting pregnant but usually are fertile. Thicker cervical mucus makes it harder for sperm to penetrate the cervix. In some patients, the poor nutrition that accompanies CF can cause irregular ovulation, making pregnancy difficult.

Bowel obstruction may occur in CF because of meconium ileus, which is caused by excessively thick meconium.

How Do We Diagnose Cystic Fibrosis (LO #3)?

Most cases of CF are diagnosed through newborn screening, DNA genetic testing, or sweat testing.

Newborn Screening

Many nations (including the United States) now require routine screening of newborns for CF. A variety of panels that detect CFTR gene mutations are commonly used in screening. These panels have largely replaced the older test for immunoreactive trypsinogen (IRT), a pancreatic protein that is reduced in the serum of CF patients. The IRT test has many false positives, leading to its replacement by gene panels.

CLINICAL CORRELATION

Every state in the U.S. has a newborn screening program that screens newborns for many serious but treatable congenital diseases. Many of these conditions are detected by testing a small sample of blood taken from a newborn's heel. The conditions screened for include spinal muscular atrophy, cystic fibrosis, sickle cell disease and other hemoglobinopathies, endocrine diseases, inborn errors of metabolism, lysosomal storage diseases, severe combined immunodeficiencies, critical congential heart defects, and hearing loss. Since the symptoms of these serious conditions do not always appear at birth, early detection is lifesaving and enables children to reach their full potential.

Sweat Chloride Testing

If the newborn screening test is positive, or if a patient did not receive newborn screening but now has typical CF symptoms, diagnostic testing must be done to show the functional effects of the CFTR mutation. This is done by the sweat chloride test (officially termed the quantitative pilocarpine iontophoresis sweat test [QPIT]), which analyzes a sample of patient sweat for chloride content.

How does sweat testing work? A skin effect of CF is that the mutated CFTR prevents absorption of chloride (and indirectly sodium) into the epithelial cells lining the sweat ducts (Figure 3). This causes **increased chloride and sodium in the sweat**.

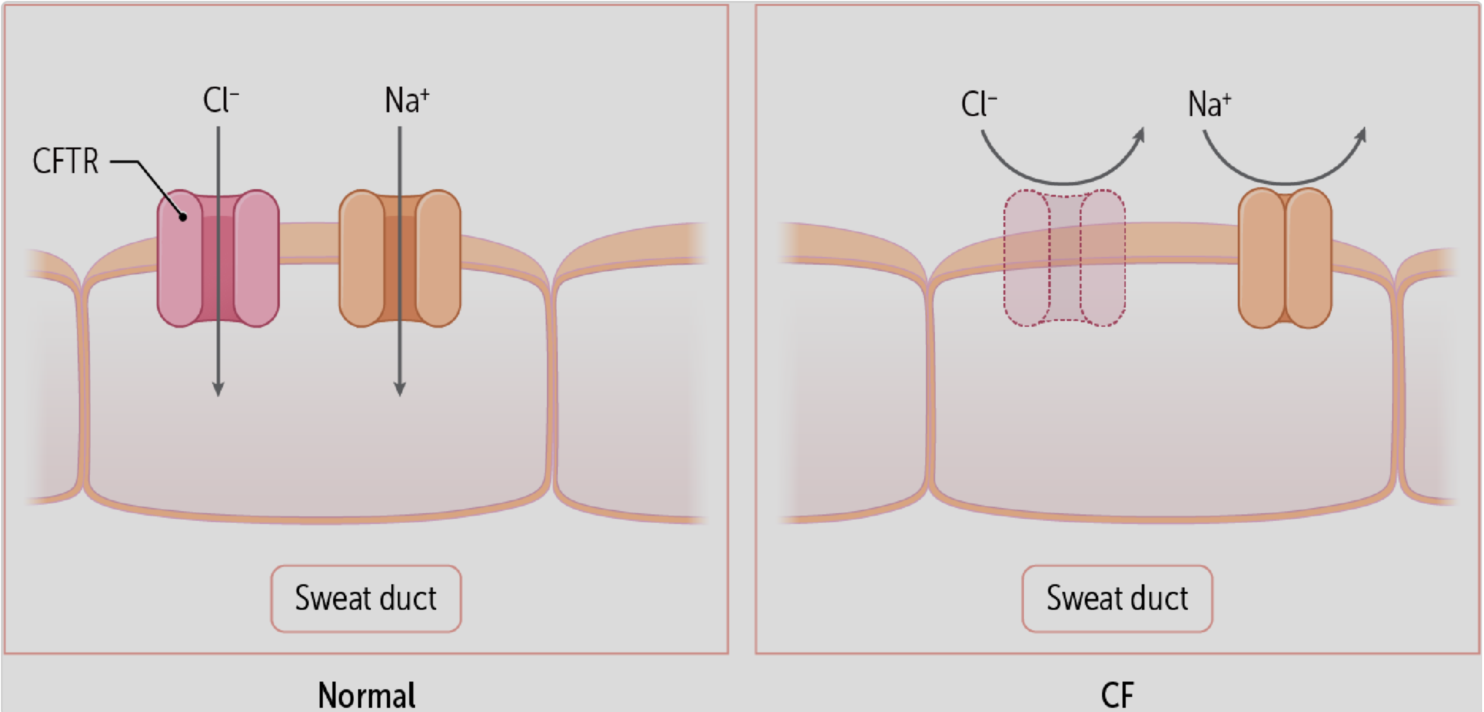

Figure 3

The test is performed by delivering pilocarpine, an M3 muscarinic agonist, into a small area on the forearm. This stimulates sweat production. Sweat from the stimulated area is collected and analyzed for chloride content.

Newborn screening for CF uses DNA mutation panels or immunoreactive trypsinogen testing.

How Do We Manage Cystic Fibrosis?

There is no known cure for cystic fibrosis. Nonpharmacologic and pharmacologic methods are used to reduce symptoms and complications, especially infections. Treatment of CF is covered in depth in the Cystic Fibrosis (2 of 2) Brick.

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to TF, what is the potential cause of her hyperglycemia?

You tell TF, "Just as the excess mucus plugs your breathing tubes and causes lung problems, it can also plug the ducts in your pancreas, the organ that produces insulin. Over time, this mucus plugging can also damage the pancreatic cells that secrete insulin, the hormone that regulates blood sugar. Because you don't make enough insulin, your blood sugar goes up, causing your new symptoms. This causes a special type of diabetes called cystic fibrosis–related diabetes. We can treat it just as in other forms of diabetes with insulin injections. We need to maintain your high-calorie diet because of your other CF complications, so don't use the diet guidelines you will find online for regular diabetes."

You send TF to a dietician and endocrinologist to refine the management plan.

Summary

- Cystic fibrosis (CF) is a multisystemic genetic disorder causing reduced chloride transport through epithelial cells, leading to thickened secretions, including thick airway mucus and obstructive respiratory disease.
- The white population with European ancestry has the highest prevalence of CF.
- Symptoms include wheezing, dyspnea, productive cough, failure to thrive, malabsorption, weight loss, and infertility. Patients have frequent recurrent respiratory infections.
- Mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene cause abnormal chloride transport in epithelial cells.
- The mutations decrease the amount of chloride secreted from the cell, causing water to enter the cell and thickening the extracellular fluids (including mucus and pancreatic fluids). The increased fluid thickness causes obstruction.
- Respiratory consequences include obstructive pulmonary disease, bronchiectasis, sinus obstruction, and sinus/lung infections.
- Gastrointestinal consequences include constipation, meconium ileus in neonates, malabsorption, secondary diabetes mellitus, and biliary/liver disease.
- Reproductive consequences include difficulty conceiving in females and infertility in males.
- Cystic fibrosis is usually diagnosed in prenatal or neonatal screening. CF is screened for in all newborns using IRT and DNA genetic tests.
- Sweat chloride testing confirms the diagnosis, with increased sweat chloride indicating CF.

Review Questions

1. **A mother brings her 4-day-old baby to the clinic because of poor feeding and respiratory distress. The mother notes the baby has not yet passed stool but is urinating regularly. Examination reveals that the baby is breathing rapidly, using accessory respiratory muscles for breathing. The mother says the baby was born at home, and no newborn screening was performed. Which of the following conditions, if diagnosed, would be most specific for a diagnosis of cystic fibrosis (CF) in this infant?**

   A. Bronchopulmonary dysplasia

   B. Cyanosis

   C. Intussusception

   D. Irritability

   E. Meconium ileus

Explanation (requires correct answer)

2. **Which of the following is the best test to confirm a diagnosis of cystic fibrosis (CF)?**

   A. Immunoreactive trypsinogen

   B. Methacholine challenge

   C. Oral glucose tolerance test

   D. Quantitative pilocarpine iontophoresis sweat test

   E. Spirometry

Explanation **(requires correct answer)**

A. It is an autosomal recessive disease.

B. It is caused by a mutation in the cystic fibrosis transmembrane conductance regulator (CFTR) gene.

C. Reduced epithelial cell chloride transport leads to the production of excessively thick mucus.

D. Some patients with CF produce functional CFTR but not enough of it.

E. The CFTR gene mutation directly inhibits an epithelial sodium transporter.

Explanation (requires correct answer)

How would you rate this Brick?

References

- Katkin JP. Cystic Fibrosis: clinical manifestations and diagnosis. In: UpToDate, Hoppin AG (Ed), Wolters Kluwer. (Accessed on April 27, 2025.)
- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 **Differentiate the clinical presentations of cystic fibrosis (CF).**

02 **Explain the pathophysiology, including the genetic basis, of CF.**

03 **Interpret clinical findings to properly diagnose CF.**

Objective Confidence: 0/3